

Overview of Anomaly detection

Subject: Anomaly detection

1.

Parameter-free Also referred to as frequency-based or counting-based, the simplest non-parametric anomaly detection method is to build a histogram with the training data or a set of known normal instances, and if a test point does not fall in any of the histogram bins mark it as anomalous, or assign an anomaly score to test data based on the height of the bin it falls in. The size of bins are key to the effectiveness of this technique but must be determined by the implementer. A more sophisticated technique uses kernel functions to approximate the distribution of the normal data. Instances in low probability areas of the distribution are then considered anomalies.

2.

The Subspace Outlier Degree (SOD) identifies attributes where a sample is normal, and attributes in which the sample deviates from the expected. **Correlation Outlier Probabilities (COP)** compute an error vector of how a sample point deviates from an expected location, which can be interpreted as a counterfactual explanation: the sample would be normal if it were moved to that location.

3.

Anomaly detection in dynamic networks Dynamic networks, such as those representing financial systems, social media interactions, and transportation infrastructure, are subject to constant change, making anomaly detection within them a complex task. Unlike static graphs, dynamic networks reflect evolving relationships and states, requiring adaptive techniques for anomaly detection.

4.

Ill defined An outlier is an observation which deviates so much from the other observations as to arouse suspicions that it was generated by a different mechanism. Anomalies are instances or collections of data that occur very rarely in the data set and whose features differ significantly from most of the data. An outlier is an observation (or subset of observations) which appears to be inconsistent with the remainder of that set of data. An anomaly is a point or collection of points that is relatively distant from other points in multi-dimensional space of features. Anomalies are patterns in data that do not conform to a well-defined notion of normal behaviour.

5.

Oil and gas pipeline monitoring In the oil and gas sector, anomaly detection is not just crucial for maintenance and safety, but also for environmental protection. Aljameel et al. propose an advanced machine learning-based model for detecting minor leaks in oil and gas pipelines, a task traditional methods may miss.

6.

Methods Many anomaly detection techniques have been proposed in literature. The performance of methods usually depend on the data sets. For example, some may be suited to detecting local outliers, while others global, and methods have little systematic advantages over another when compared across

many data sets. Almost all algorithms also require the setting of non-intuitive parameters critical for performance, and usually unknown before application. Some of the popular techniques are mentioned below and are broken down into categories:

7.

Intrusion detection Anomaly detection was proposed for intrusion detection systems (IDS) by Dorothy Denning in 1986. Anomaly detection for IDS is normally accomplished with thresholds and statistics, but can also be done with soft computing, and inductive learning. Types of features proposed by 1999 included profiles of users, workstations, networks, remote hosts, groups of users, and programs based on frequencies, means, variances, covariances, and standard deviations. The counterpart of anomaly detection in intrusion detection is misuse detection.

8.

Specific Let T be observations from a univariate Gaussian distribution and O a point from T . Then the z -score for O is greater than a pre-selected threshold if and only if O is an outlier.

9.

Explainable anomaly detection Many of the methods discussed above only yield an anomaly score prediction, which often can be explained to users as the point being in a region of low data density (or relatively low density compared to the neighbor's densities). In explainable artificial intelligence, the users demand methods with higher explainability. Some methods allow for more detailed explanations:

10.

Quantum Quantum machine learning approaches have been investigated for anomaly detection on near-term quantum hardware. Quantum support vector machines (QSVMs) using quantum kernel methods, such as the ZZFeatureMap, have been applied to classification of normal versus anomalous instances by leveraging high-dimensional quantum feature spaces.

11.

Definition Many attempts have been made in the statistical and computer science communities to define an anomaly. The most prevalent ones include the following, and can be categorised into three groups: those that are ambiguous, those that are specific to a method with pre-defined thresholds usually chosen empirically, and those that are formally defined:

12.

Intrusion detection The concept of intrusion detection, a critical component of anomaly detection, has evolved significantly over time. Initially, it was a manual process where system administrators would monitor for unusual activities, such as a vacationing user's account being accessed or unexpected printer activity. This approach was not scalable and was soon superseded by the analysis of audit logs and system logs for signs of malicious behavior. By the late 1970s and early 1980s, the analysis of these logs was primarily used retrospectively to investigate incidents, as the volume of data made it impractical for real-time monitoring. The affordability of digital storage eventually led to audit logs being analyzed online, with specialized programs being developed to sift through the data. These programs, however, were typically run during off-peak hours due to their computational intensity. The

1990s brought the advent of real-time intrusion detection systems capable of analyzing audit data as it was generated, allowing for immediate detection of and response to attacks. This marked a significant shift towards proactive intrusion detection. As the field has continued to develop, the focus has shifted to creating solutions that can be efficiently implemented across large and complex network environments, adapting to the ever-growing variety of security threats and the dynamic nature of modern computing infrastructures.

13.

Software ELKI is an open-source Java data mining toolkit that contains several anomaly detection algorithms, as well as index acceleration for them. PyOD is an open-source Python library developed specifically for anomaly detection. scikit-learn is an open-source Python library that contains some algorithms for unsupervised anomaly detection. Wolfram Mathematica provides functionality for unsupervised anomaly detection across multiple data types